Version 1

1. Red light has a wavelength of 680 nm while green light has a wavelength of 540 nm. What does this information tell you about the relative frequencies and energies of red and green light?

a. The frequency of red light is = lower than green light

b. The energy of red light is = lower than green light

*wavelength and frequency are inversely proportional

2. Light that has a frequency of $7.14 \times 10^{14}\ s^{-1}$ appears violet to humans. What is the wavelength of this light in units of nm?

a. 4.20×10^{-7} nm
b. 2.14×10^{23} nm
c. 2.14×10^{14} nm
d. 2.38×10^{-3} nm
e. 420 nm

$c = \lambda\nu$

$3.00 \times 10^{8} \frac{m}{s} = \lambda (7.14 \times 10^{14} s^{-1})$

$4.20 \times 10^{-7} m = \lambda$

$4.20 \times 10^{-7} m \left(\frac{10^{9} nm}{1 m}\right) = 420 nm$

*But answers are listed in nm.

3. The energy of a photon that has a wavelength of 12.3 nm is _____ J.

a. 1.51×10^{-17}
b. 4.42×10^{-23}
c. 1.99×10^{-25}
d. 2.72×10^{-50}
e. 1.62×10^{-17}

$E = nh\nu$ ← need frequency → $c = \lambda\nu$

$E = (1)(6.626 \times 10^{-34} J \cdot s)(2.439 \times 10^{16} s^{-1})$

$3.00 \times 10^{8} \frac{nm}{s} = (12.3 nm)\left(\frac{1 m}{10^{9} nm}\right)\nu$

of photons

$2.439 \times 10^{16} s^{-1} = \nu$

$E = 1.62 \times 10^{-17} J$

4. CO_2 strongly absorbs radiation re-emitted from earth with a frequency of 2.00×10^{14} Hz. How much energy, in kJ, is associated with 1.00 mol of photons of radiation with this frequency?

$1 Hz = 1 s^{-1}$

a. 5.99×10^{-19} kJ
b. 1.20×10^{35} kJ
c. 1.33 kJ
d. 79.8 kJ
e. 32.7 kJ

$E = nh\nu$

$E = (6.022 \times 10^{23} photons)(6.626 \times 10^{-34} J \cdot s)(2.00 \times 10^{14} s^{-1})$

1 mol photons

$E = 7.98 \times 10^{5} J \left(\frac{1 kJ}{1000 J}\right) = 79.8 kJ$

5. The phenomenon of "line spectra" led to the discovery that:
a. matter has wavelike properties
b. energy has particle like properties
c. energy states of the electron are quantized
d. electronic transitions always release visible light
e. radiant energy is quantized

6. What wavelength of light is emitted as an electron in the hydrogen atom falls from n = 5 to n = 3?
a. 1.28×10^{-6} m
b. 1.55×10^{-19} m
c. 2.34×10^{14} m
d. 7.02×10^{22} m
e. 0.0711 m

$\Delta E = (R_H)\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$

$\Delta E = (2.18 \times 10^{-18}\ J)\left(\frac{1}{5^2} - \frac{1}{3^2}\right)$

$\Delta E = -1.55 \times 10^{-19}\ J$

negative sign indicates energy was released

$E = h\nu$

$1.55 \times 10^{-19}\ J = (6.626 \times 10^{-34}\ J \cdot s)\nu$

$2.339 \times 10^{14}\ s^{-1} = \nu$

$c = \lambda\nu$

$3.00 \times 10^{8}\ m/s = \lambda(2.339 \times 10^{14}\ s^{-1})$

$1.28 \times 10^{-6}\ m = \lambda$

7. What is the de Broglie wavelength of an oxygen molecule (O_2) molecule moving at 478 m/s? (1 amu = 1.66×10^{-24} g)
a. 2.61×10^{-11} m
b. 2.61×10^{-14} m
c. 4.33×10^{-38} m
d. 4.33×10^{-35} m
e. 3.83×10^{10} m

$\lambda = \frac{h}{mv}$ ← has units of J·s

m = mass, v = velocity

* Remember $1\ J = 1\ \frac{kg \cdot m^2}{s^2}$

so mass needs to be in kg for units to cancel

$$\lambda = \frac{6.626 \times 10^{-34}\ J \cdot s}{(31.998\ amu)\left(\frac{1.66 \times 10^{-24}\ g}{1\ amu}\right)\left(\frac{1\ kg}{1000\ g}\right)\left(478\ \frac{m}{s}\right)}$$

(31.998 amu = mass of O_2 molecule)

$= 2.61 \times 10^{-11}\ m$

8. A particular orbital is designated $5d_{yz}$. What do each of the components of the name describe about the orbital?

a. 5 = energy state

b. d = shape

c. yz = spatial orientation

9. How many ***allowed values*** are there for the quantum number, "*l*", when n = 3?

a. 0
b. 1
c. 2
d. 3
e. 4

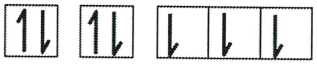
allowed values for ℓ → $0, \ldots, n-1$

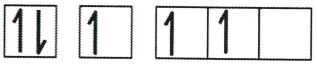
so 0, 1, 2 are the allowed values

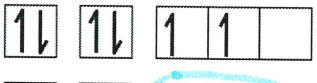
3 allowed values for ℓ

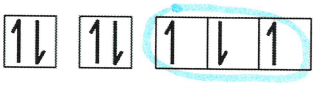
10. Shown below are some allowed and forbidden electron configurations for the highest energy s and p orbitals of atoms. Classify the electron configuration as follows:
Ground state
An allowed excited state
Violation of Pauli exclusion principle
Violation of Hund's rule

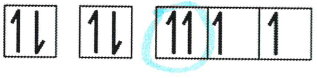
a.  violation of Pauli ex. principle - no two electrons can have the same 4 quantum numbers

b.  ground state

c.  allowed excited state

d.  ground state

e.  violation of Hund's Rule: electrons with same spin maximized in degenerate orbitals

f.  violation of Pauli Ex. Prin.

a. a =

b. b =

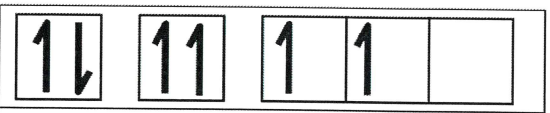
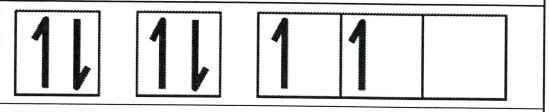
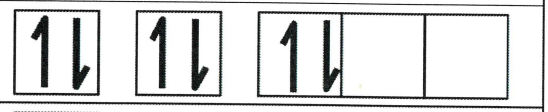
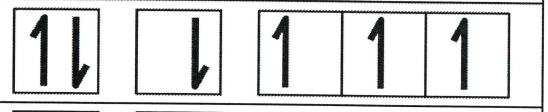
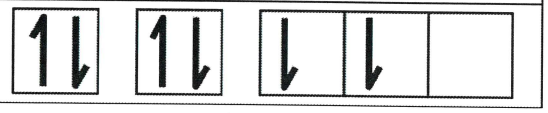
c. c =

d. d =

e. e =

f. f =

11. Which of the electron configurations represents an allowed excited state?

(i)		Violates Pauli Ex. Prin.
(ii)		ground
(iii)		violates Hund's Rule
(iv)		
(v)		ground

a. (i)
b. (ii)
c. (iii)
d. (iv)
e. (v)

12. Write the electron configuration for technetium (Tc) abbreviating with the appropriate noble gas core.

a. $[Kr]5s^2 4d^5$
b. $[Xe]5s^2 4d^5$
c. $[Kr]5s^2 5d^5$
d. $[Kr]5s^2 6d^5$
e. $[Xe]\ 4s^2 4d^5$

Tc: $[Kr]\ 5s^2 4d^5$

↑ 1st 36 electrons

*Use Aufbau diagram to predict the order of orbital fillings.

13. Which of the following elements would contain unpaired electrons in the ground state? Select all that apply, this is a multiple response question.

a. K : [Ar] $\frac{1}{4s}$

b. Ca : [Ar] $\frac{1\downarrow}{4s}$

c. Ti : ⟶ [Ar] $\frac{1\downarrow}{4s}$ $\frac{1}{3d}$ $\frac{1}{3d}$ $\frac{\ }{3d}$ $\frac{\ }{3d}$

d. Cd : [Kr] $\frac{1\downarrow}{5s}$ $\frac{1\downarrow 1\downarrow 1\downarrow 1\downarrow 1\downarrow}{4d}$

e. As: ⟶ [Ar] $\frac{1\downarrow}{4s}$ $\frac{1\downarrow 1\downarrow 1\downarrow 1\downarrow 1\downarrow}{3d}$ $\frac{1\downarrow}{4p}$ $\frac{1}{4p}$ $\frac{1}{4p}$

f. Ne $\frac{1\downarrow}{1s}$ $\frac{1\downarrow}{2s}$ $\frac{1\downarrow 1\downarrow 1\downarrow}{2p}$

g. Ag:
[Kr] $\frac{1}{5s}$ $\frac{1\downarrow 1\downarrow 1\downarrow 1\downarrow 1\downarrow}{4d}$

*Even if you forgot the anomalous configuration, it would still have unpaired electrons.

14. Which of the following elements would you expect to have anomalous electron configurations? Select all that apply, this is a multiple response question.

a. Zn – No (4s and 3d orbitals completely filled)

b. Cr – Yes [Ar] $\underset{4s}{\uparrow}$ $\underset{3d}{\uparrow\uparrow\uparrow\uparrow\uparrow}$

c. Ag – Yes [Kr] $\underset{5s}{\uparrow}$ $\underset{4d}{\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow}$

d. Sc – No [Ar] $\underset{4s}{\uparrow\downarrow}$ $\underset{3d}{\uparrow}$ ___ ___ ___

15. How many electrons are capable of occupying the (i) n = 2, (ii) n = 3 and (iii) n = 4 shells?

a. (i) 2;	(ii) 8;	(iii) 18
b. (i) 16;	(ii) 32;	(iii) 64
c. (i) 4;	(ii) 6;	(iii) 8
d. (i) 4;	(ii) 8;	(iii) 16
e. (i) 8;	(ii) 18;	(iii) 32

(i) n=2 (ones, three p orbitals) – total of eight electrons

(ii) n=3 (ones, three p, five d orbitals) – total of eighteen electrons

(iii) n=4 (ones, three p, five d, seven f orbitals) – total of thirty-two electrons

16. When determining the effective nuclear charge of an electron in an atom in a row of the periodic table using the simple rule $Z_{eff} = Z - S$, what assumptions must be made?

a. (a) all electrons of a given "n" value are treated identically – yes

b. (b) electrons with the same "n" value do not shield one another significantly – yes

c. (c) electrons in energy levels higher than the energy level of the electron of interest provide shielding – no

d. (d) both assumptions (a) and (b) are made

e. all of the above assumptions are made

17. The electrons in which subshell of selenium (Se) will feel the least effective nuclear charge?

a. 2p

b. 3p

c. 3d

d. 4p – Electrons in the 4p orbital will be shielded by electrons in the n=1, 2, and 3 shells and even slightly by the 4s electrons.

e. 4s

18. For which of the following atoms are the n = 2 electrons closest to the nucleus?

for n=2 e⁻

a. F — $Z_{eff} = 9 - 2 = +7$

b. Cl — $Z_{eff} = 17 - 2 = +15$

c. Br — $Z_{eff} = 35 - 2 = +33$

d. I — $Z_{eff} = 53 - 2 = +51$

[e.] At — $Z_{eff} = 85 - 2 = +83$

↳ only the 2 electrons in the n=1 shell will shield electrons in the n=2 shell.

$Z_{eff} = Z - S$

↑ # protons

e⁻ in n=2 of At will be closest to the nucleus

19. Order the following atoms from ***smallest to largest*** radius: Ar, He, Na, Si

a. He < Si < Na < Ar

b. Na < Si < Ar < He

[c.] He < Ar < Si < Na

d. Si < Ar < Na < He

e. Ar < Ne < Si < He

← atomic radius trend ↓

20. Which ***two*** (i.e. select two answers) of the following elements in n = 3 would you predict would have ionization energy values that do not follow the general trend for how values change when moving from left to right in a row?

a. Na

b. Mg

[c.] Al – due to shielding of "s" electrons

d. Si

e. P

[f.] S – due to minimizing e⁻ pair repulsion

g. Cl

h. Ar

21. For which ionization of B would you expect the ionization energy to increase most dramatically from the previous value?

a. 2nd

b. 3rd

[c.] 4th – would require removal of e⁻ from n=1 core

d. 5th

22. The general trend in electron affinity (with a few notable exceptions) for s and p block elements in the periodic table is that electron affinity becomes more negative when moving from ___(i)___ across a row and moving from ___(ii)___.

a. (i) left to right (ii) top to bottom
b. (i) left to right (ii) bottom to top
c. (i) left to right (ii) no general trend in the vertical direction
d. (i) right to left (ii) top to bottom
e. (i) right to left (ii) bottom to top
f. (i) right to left (ii) no general trend in the vertical direction

23. The ionization energy for an isolated gaseous atom of potassium is 419 kJ/mol. What is the longest wavelength of electromagnetic radiation capable of ionizing sodium atoms in the gaseous state?

a. 2.86×10^{-4} m
b. 2.86×10^{-7} m
c. 1.05×10^{15} m
d. 1.51×10^{36} m
e. 6.96×10^{-19} m

$E = nh\nu$

$\frac{419\ kJ}{mol} \left(\frac{1000\ J}{1\ kJ}\right)\left(\frac{1\ mol}{6.022 \times 10^{23}\ Na\ atoms}\right) = 6.958 \times 10^{-19}\ J$ per sodium atom

$6.958 \times 10^{-19}\ J = (1)(6.626 \times 10^{-34}\ J \cdot s)\nu$

$1.05 \times 10^{15}\ s^{-1} = \nu$

$c = \lambda\nu$

$3.00 \times 10^{8}\ \frac{m}{s} = \lambda\,(1.05 \times 10^{15}\ s^{-1})$

$2.86 \times 10^{-7}\ m = \lambda$

24. A decrease in first ionization energy upon going from Na to K can be attributed to what factor?

a. electron pairing within an orbital
b. electron-nuclear distance — 4s electron in K farther from nucleus than 3s e⁻ in Na
c. effective nuclear charge
d. electron-electron repulsion between core and valence electrons

25. Order the following atoms from highest to lowest first-ionization energy: Cl, S, Se, Sb

a. S > Cl > Se > Sb
b. Cl > Se > S > Sb
c. Cl > S > Se > Sb
d. Sb > Se > S > Cl

